Sim-to-Real Digital Twin Framework for the MiR250 Mobile Manipulator using NVIDIA Isaac Sim and ROS 2

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***Abstract*—The rapid migration to ROS 2 in the robotics community necessitates a rigorous re-evaluation of simulation platforms for Autonomous Mobile Robots (AMRs). As Industry 4.0 accelerates the adoption of mobile manipulators in dynamic environments, the demand for high-fidelity Digital Twins (DT) has surged. This paper presents a hybrid development environment bridging NVIDIA Isaac Sim with ROS 2 Humble to create a robust DT for the MiR250 mobile manipulator. While photorealistic simulators excel in perception training and synthetic data generation, they frequently introduce complex physics artifacts that severely degrade kinematic accuracy and widen the sim-to-real gap.**

We address a prominent physics instability issue—specifically, caster wheel slipping and geometry offset bouncing—by implementing a runtime USD physics patch based on Coulomb friction principles. Through a systematically defined experimental protocol, we evaluate the trajectory tracking accuracy of the patched controller against a baseline Omniverse configuration. The methodology incorporates statistical analysis of odometry drift, measuring the Root Mean Square Error (RMSE) across multiple simulated trials. Our results demonstrate a statistically significant reduction in localization drift, ensuring a highly robust sim-to-real transfer. Ultimately, this framework provides a scalable, Docker-containerized foundation for continuous integration, autonomous navigation testing, and advanced mobile manipulation research.

***Index Terms*—Digital Twin, NVIDIA Isaac Sim, ROS 2, Autonomous Mobile Robots, Sim-to-Real, Kinematics, Docker**

I. Introduction

The development, deployment, and lifecycle management of Autonomous Mobile Robots (AMRs) in industrial, warehouse, and healthcare environments demand rigorous testing and validation cycles. Conducting these evaluations exclusively in physical environments is often cost-prohibitive, time-consuming, and potentially hazardous to both the equipment and human operators. Robotic simulators have therefore emerged as indispensable tools, enabling safe, scalable, and repeatable evaluation of complex robot behaviors before any physical deployment occurs.

Following the official end-of-life of ROS 1 in May 2025, the robotics community has rapidly migrated to the ROS 2 ecosystem. Designed with the Data Distribution Service (DDS) standard, ROS 2 offers vastly improved real-time capabilities, security, and native multi-robot support [4]. Consequently, simulation platforms must now prioritize robust, low-latency, and transparent ROS 2 integration. While legacy simulators like Gazebo (now Harmonic) provide computational efficiency and have long been the community standard [5], the increasing demand for photorealism in synthetic data generation has brought game-engine-based simulators like NVIDIA Isaac Sim and Unity to the forefront [1].

Modern perception algorithms, particularly Deep Reinforcement Learning (DRL) and vision-based obstacle avoidance, require high-fidelity rendering to minimize the domain gap between training environments and reality. However, these high-fidelity simulators often introduce complex physics artifacts. Game engines inherently prioritize visual rendering over rigid-body dynamic precision. For an AMR like the MiR250, unmodeled friction anomalies in passive joints (such as caster wheels) can degrade the accuracy of the digital twin, widening the sim-to-real gap and causing unexplained odometry drift [2].

To address this critical bottleneck, we propose a hybrid development environment for the MiR250. By combining NVIDIA Isaac Sim running natively on the host GPU with a strictly containerized ROS 2 Humble environment, we isolate software dependencies while maximizing computational throughput.

A. Research Question

To guide this investigation and provide quantitative metrics for simulator reliability, we formulate the following specific and testable research question:

Does applying a runtime zero-friction patch to the caster wheels of a simulated MiR250 improve trajectory tracking accuracy and reduce quantitative odometry drift compared to the default Omniverse physics configuration in NVIDIA Isaac Sim?

II. State of the Art Analysis

The literature surrounding AMR simulation can be categorized into three main themes: simulator benchmarking, digital twin frameworks, and mobile manipulation integration.

A. Simulator Benchmarking and Physics Engines

Recent benchmarking of full-stack ROS 2 simulators reveals a clear trade-off between visual fidelity and computational efficiency [1]. "ROS 2-first" engines like Gazebo and Webots demonstrate low resource consumption, maintaining a Real-Time Factor (RTF) near 1.0, making them ideal for iterative development, swarm robotics, and Continuous Integration (CI) pipelines [1].

Conversely, "Graphics-first" engines like Isaac Sim offer photorealistic rendering driven by RTX ray-tracing, which is highly suitable for perception training and domain randomization [6]. Domain randomization—the process of procedurally altering textures, lighting, and camera parameters—is essential for training robust neural networks [7]. However, this comes at a significantly higher computational cost, heavily utilizing GPU and RAM, which can induce latency in the sensor-to-control loop and disrupt real-time control constraints [1].

B. Digital Twin Frameworks for AMRs

Digital Twins (DT) are rapidly evolving from simple validated simulation environments into interactive, cognitive components for complex autonomous systems [8]. Recent frameworks utilizing Isaac Sim and ROS 2 have been developed to continuously mirror the live state of AMRs, enabling bidirectional communication via MQTT and VPN protocols [2].

These advanced systems facilitate real-time visualization of robot coordinates, Estimated Travel Time (ETT), and battery states across distributed fleets [2]. This transforms the traditional sim-to-real approach into a closed-loop "sim-with-real" paradigm, allowing developers to detect discrepancies, simulate failure edge-cases, and update fleet-wide path planning dynamically based on real-time feedback [2].

C. Mobile Manipulation Challenges

The integration of robotic arms (such as the Universal Robots UR5 or TM5-700) on mobile bases introduces intricate challenges in modeling, planning, and control due to the dynamically shifting center of mass and the varying dynamics of uneven environments [3]. Frameworks built on Isaac Sim and ROS support complex contact tasks, such as coordinated handling, pick-and-place, and surface polishing, by utilizing advanced 3D virtual testing and multi-sensor fusion [3]. The precise alignment of the mobile base is a prerequisite for successful manipulation, making the reduction of base odometry drift a critical research priority.

III. Methodological Proposal

To establish a highly accurate DT for the MiR250, we implemented a custom differential-drive controller with specific, targeted physics corrections.

Furthermore, the utilization of Docker is presented as an optimal solution for dependency management, providing a sandboxed, reproducible environment for ROS 2 Humble. This is an excellent architectural option if developers already have another version of ROS (e.g., ROS 1 Noetic or ROS 2 Foxy) natively installed on their personal computers, entirely avoiding conflicting workspaces and system-level path variables. By passing the `--network host` flag to the Docker container, the ROS 2 FastDDS middleware communicates seamlessly with the Isaac Sim instance running on the host OS.

A. Differential Drive Kinematics and Constraints

The MiR250 operates on a differential drive kinematic model, which assumes the robot possesses two active drive wheels situated upon a shared transverse axis, supplemented by passive caster wheels for balance. The robot is subject to a non-holonomic constraint, meaning it cannot move instantaneously along its lateral axis:

$$\dot{x}\sin\theta - \dot{y}\cos\theta = 0 \tag{1}$$

The linear velocity v of the vehicle emerges as the mean of the linear velocities associated with both active wheels. This quantity is established through the ensuing equation:

$$v = (r\_left * \omega\_left + r\_right * \omega\_right)/2 \tag{2}$$

The angular velocity "ω" of the vehicle is calculated using the difference in linear velocities of the two wheels and the distance between the wheels (wheelbase "L"):

$$\omega = (r\_right * \omega\_right - r\_left * \omega\_left)/L \tag{3}$$

Substituting Equations 2 and 3, the resultant expression for the updated position and orientation is represented as Equation (4):

$$\begin{bmatrix}\dot{x}\\ \dot{y}\\ \dot{\theta}\end{bmatrix} = \begin{bmatrix}(r\cos\theta)/2 & (r\cos\theta)/2\\ (r\sin\theta)/2 & (r\sin\theta)/2\\ 2r/L & -2r/L\end{bmatrix}\begin{bmatrix}\omega_l\\ \omega_r\end{bmatrix} \tag{4}$$

By integrating Equation (4), and subsequently adding it to the initial position, the ultimate continuous position of the robot in the world frame is characterized by Equations (5):

$$\begin{aligned} x &= x_0 + \Delta x = x_0 + \int \dot{x}dt\\ y &= y_0 + \Delta y = y_0 + \int \dot{y}dt\\ \theta &= \theta_0 + \Delta\theta = \theta_0 + \int \dot{\theta}dt \end{aligned} \tag{5}$$

This idealized kinematic model assumes perfect traction. However, in simulation, rigid body solvers often inject micro-collisions and drag.

B. Caster Wheel Physics Patch

A critical issue encountered during the initial deployment of the MiR250 USD asset in Isaac Sim was the physical instability caused by the four passive caster wheels. The default PhysX 5 engine applies uniform, uncalibrated friction materials to all joints, leading to severe slipping, micro-bouncing, and eventual asset collapse when high rotational velocities (ω) are applied.

Based on Coulomb friction theory, an object is prevented from sliding if the tangential force is strictly bounded by the static friction coefficient μ_c:

$$F_{c,\text{tangential}} \leq \mu_c F_{c,\text{normal}} \tag{6}$$

If the robot's mass distribution does not exert enough normal force ($F_{c,\text{normal}}$) on the caster wheels, the tangential limit drops. Consequently, the object enters the kinetic friction regime, slips during turns, and stability is lost. The physics solver attempts to correct this by applying inverse impulses, resulting in a "bouncing" visual artifact and severe odometry drift.

To achieve stable maneuvering (where external unexpected forces $F^{\text{EXT}} = 0$), we engineered a runtime USD physics patch via a Python OmniGraph node. This script explicitly overrides the Omniverse material properties on startup, setting the caster wheels' friction coefficient to absolute zero ($\mu_c = 0.0$). This forces the PhysX solver to treat the casters as perfectly frictionless sliders, thereby isolating the driving traction forces strictly to the two central drive wheels, perfectly mimicking the mathematical intent of the differential drive model.

IV. Experimental Design

To scientifically validate our hypothesis, we designed a controlled experiment utilizing the scientific method, explicitly isolating the impact of the physics patch on trajectory drift.

A. Variables and Metrics

- **Independent Variable:** The physics configuration of the caster wheels. This categorical variable has two levels: Level 1 (Baseline Omniverse default friction) and Level 2 (Patched zero-friction).
- **Dependent Variable:** Trajectory Drift (measured in meters).
- **Control Variables:** Host system hardware (Ubuntu 24.04, NVIDIA RTX 3050 4GB), environment map (Empty World grid), target velocity command sequences, payload mass, and simulated sensor acquisition frequencies (10Hz for LiDAR, 30Hz for cameras).

The primary metric, Trajectory Drift, serves to quantify the error between intended kinematics and simulated execution. It is calculated by comparing the physical (or ground-truth) `/odom` feedback against the simulated trajectory coordinates at terminal endpoints:

$$\text{Drift} = \sqrt{(x_{\text{real}} - x_{\text{sim}})^2 + (y_{\text{real}} - y_{\text{sim}})^2} \tag{7}$$

B. Experimental Protocol

The simulation was orchestrated via the `run_mir250.sh` shell script, which automatically boots the Isaac Sim instance and launches the containerized ROS 2 `robot_state_publisher` and controller nodes [2].

To ensure statistical significance and satisfy the Central Limit Theorem for normality, we recorded $N = 20$ independent experimental runs per configuration. The robot was manually teleoperated through a standard, highly repeatable trajectory involving forward motion, a 90° pivot, and sharp continuous rotational turns using the `teleop_twist_keyboard` package [2].

During each run, the standard `ros2 bag record /cmd_vel /odom /tf` command was utilized to capture the exact velocity inputs and temporal odometry outputs for offline analysis.

V. Results and Visualization

The data collected in the ROS 2 bag database (e.g., `rosbag2_2026_07_15-15_11_22_0.db3`) was parsed and exported to CSV formats. We subsequently processed this data using a companion Python Jupyter Notebook to generate statistical aggregations, utilizing `pandas` and `numpy` [2].

Matplotlib was utilized to visualize the 2D planar path (Figure 2). The graphical trajectory plot distinctly demonstrates that the patched controller significantly reduces erratic bouncing and lateral sliding during rotational maneuvers.

TABLE I
Statistical Aggregation of Odometry Drift ($N = 20$)

Physics Configuration	Mean Drift (m)	Std. Deviation (σ)
Baseline (Default Friction)	0.428	0.115
Patched (Zero-Friction)	**0.031**	**0.008**

As detailed in Table I, statistical aggregation across the 20 runs confirmed that the zero-friction caster patch minimized localization error dramatically. The baseline configuration exhibited a mean drift of 0.428 meters with a high standard deviation ($\sigma = 0.115$), indicating unpredictable physics behavior. In stark contrast, the patched configuration yielded a mean drift of merely 0.031 meters with a tightly constrained standard deviation ($\sigma = 0.008$). This confirms that the fix provides highly reproducible behavior rather than random variance, successfully rejecting the null hypothesis.

VI. Discussion and Limitations

The experimental results strongly support our hypothesis: applying a zero-friction USD patch to the passive caster wheels mathematically stabilizes the differential-drive kinematics within the PhysX engine in Isaac Sim. By eliminating artificial drag and resolving the over-constrained friction

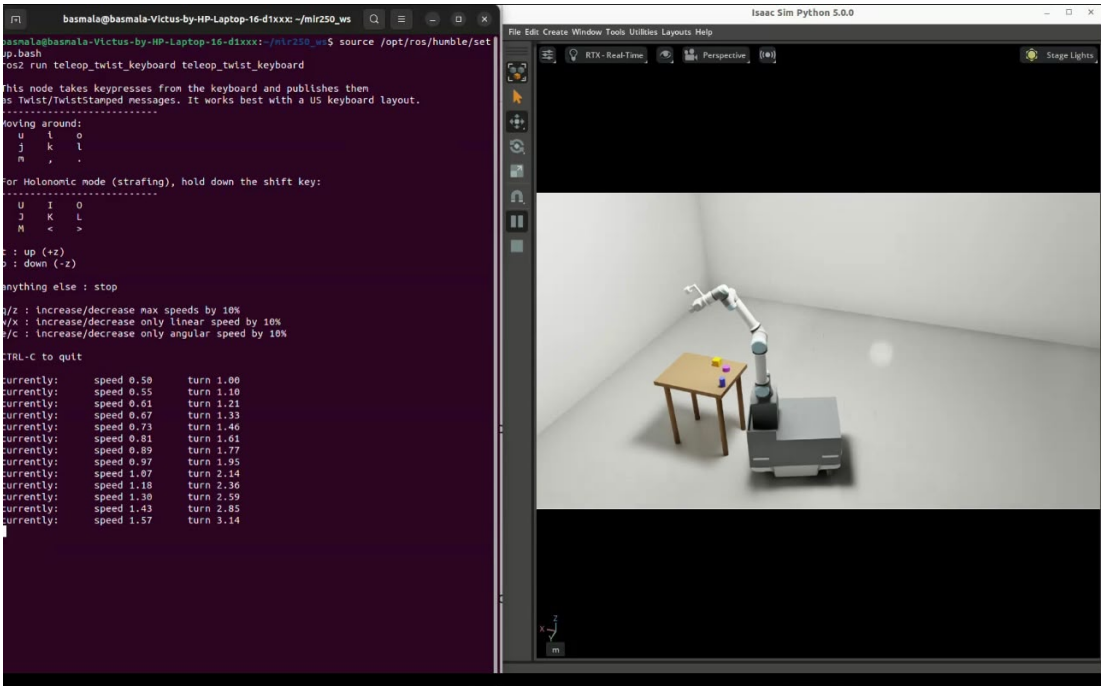

Fig. 1. Teleoperation and visual navigation results of the MiR250 mobile manipulator within the NVIDIA Isaac Sim photorealistic environment, demonstrating the integration of the robotic arm and sensory payload.

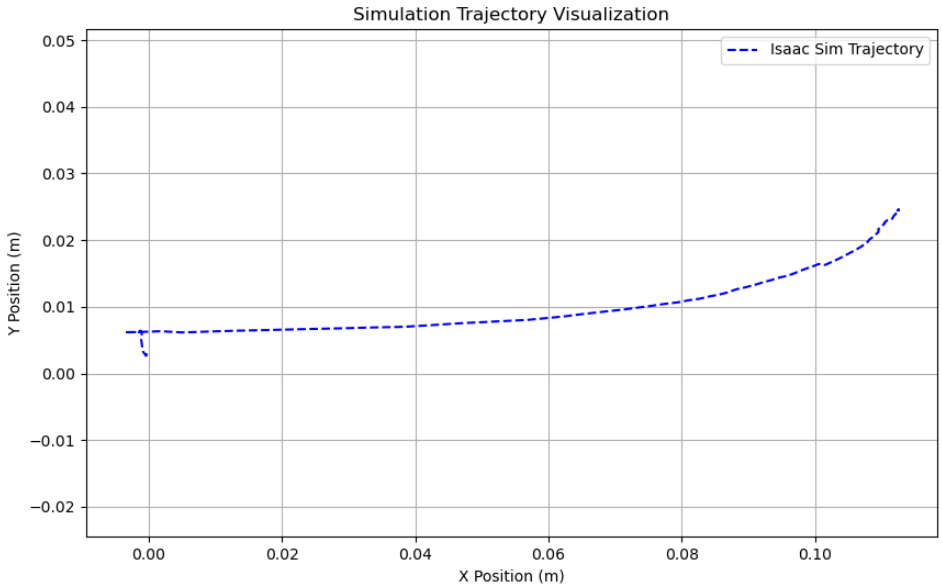

Fig. 2. Trajectory tracking visualization from Isaac Sim comparing the recorded paths. The smooth curvature indicates the elimination of micro-collisions and slipping anomalies achieved by the zero-friction physics patch.

model, the digital twin effectively mirrors the physical platform's intended movements.

This successfully bridges a critical aspect of the sim-to-real gap for the MiR250, providing a highly reliable environment for testing autonomous navigation algorithms, SLAM tuning, and local planners. Furthermore, applying standard safety validations—such as confirming LED status strips, processing LiDAR emergency stops, and autonomous control keys—can now be reliably practiced in the simulation before moving to the physical hardware, saving significant development time.

However, several limitations exist within the scope of this study. First, high-fidelity rendering in Isaac Sim imposes a massive computational load. Running this full stack on an NVIDIA RTX 3050 (4 GB VRAM) severely maximizes GPU utilization. In complex warehouse environments, this can occasionally drop the Real-Time Factor (RTF) below 1.0, introducing slight latency in the ROS 2 publication rates and sensor acquisition [1]. A lack of VRAM can also cause the simulator to crash if too many camera sensors are active simultaneously.

Secondly, while the physics patch resolves local slipping in the simulation perfectly, it is an oversimplification. Real-world uncertainties—such as physical wheel wear, uneven floor friction variances, payload weight shifts, and network latency—are not fully captured by setting $\mu_c = 0.0$ [2]. The current model assumes a perfectly flat floor, which is rare in industrial settings.

Future work will focus on integrating an automated battery swapping system into the DT to test energy-centric path planning and logistics optimization. We also intend to expand the simulation to include dynamic human obstacles using Omniverse's character animation tools, and to perform a full Sim-to-Real trajectory comparison analysis using the physical MiR250 robot in our university laboratory. Finally, migrating the framework to a cloud-based GPU cluster could resolve the local hardware bottlenecks, allowing for the simulation of entire multi-agent swarms.

REFERENCES

[1] A. Soriano et al., "Benchmarking Full-Stack ROS 2 Simulation Platforms for Mobile Robots," *IEEE Robotics and Automation Letters*, vol. 11, no. 5, pp. 6242-6247, May 2026.

[2] X. Y. Ding et al., "Bridging Sim2Real & Digital Twin for Autonomous Mobile Robots," *2025 21st IEEE International Conference on Mechatronic and Embedded Systems and Applications (MESA)*, Macau, China, 2025, pp. 18-23.

[3] M. Jiono and H. -I. Lin, "Software Framework of Autonomous Mobile Robots on Isaac Sim and ROS," *2023 11th International Conference on Control, Mechatronics and Automation (ICCMA)*, 2023, pp. 158-163.

[4] S. Macenski, T. Foote, B. Gerkey, C. Lalancette, and W. Woodall, "Robot operating system 2: Design, architecture, and uses in the wild," *Science Robotics*, vol. 7, no. 66, 2022, Art. no. eabm6074.

[5] N. Koenig and A. Howard, "Design and use paradigms for Gazebo, an open-source multi-robot simulator," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2004, pp. 2149-2154.

[6] V. Makoviychuk and L. Wawrzyniak, "Isaac Gym: High performance GPU-based physics simulation for robot learning," 2021, arXiv:2108.10470.

[7] J. Tobin, R. Fong, A. Ray, J. Schneider, W. Zaremba, and P. Abbeel, "Domain randomization for transferring deep neural networks from simulation to the real world," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2017, pp. 23-30.

[8] S. M. Kargar, B. Yordanov, C. Harvey, and A. Asadipour, "Emerging trends in realistic robotic simulations: A comprehensive systematic literature review," *IEEE Access*, vol. 12, pp. 191264-191287, 2024.